



High light intensity improves yield of specialized metabolites in medicinal cannabis (*Cannabis sativa* L.), resulting from both higher inflorescence mass and concentrations of metabolites

Wannida Sae-Tang^a, Ep Heuvelink^a, Céline C.S. Nicole^b, Elias Kaiser^a, Kjell Sneeuw^a, Mexximiliaan M.S.F. Holweg^a, Sabrina Carvalho^b, Iris F. Kappers^c, Leo F.M. Marcelis^{a,*}

^a Horticulture and Product Physiology, Wageningen University and Research, Wageningen, Netherlands

^b Philips Horticulture LED Solutions by Signify, Eindhoven, Netherlands

^c Laboratory of Plant Physiology, Wageningen University and Research, Wageningen, Netherlands

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ABSTRACT

Medicinal cannabis (*Cannabis sativa* L.) contains various plant specialized metabolites, such as cannabinoids and terpenoids. These metabolites are mainly accumulated in inflorescences and are the primary focus of cultivation. Medicinal cannabis is often cultivated in indoor farming with electrical light, which allows for light intensities to optimize quantity and quality of production. Although it is known that an increase in light intensity results in increased inflorescence yield in cannabis, its impact on specialized metabolites remains unclear. We aim to quantify the effects of light intensity on the yield of specialized metabolites, and to elucidate which plant traits explain these effects, using a yield component analysis. Hereto, we conducted an experiment in a climate-controlled chamber using broad-band white LEDs at three light intensities: 600, 800, and 1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ photosynthetic photon flux density (PPFD), applied during the generative phase (8 weeks with a 12-h photoperiod) in cannabis cv. 'Critical CBD'. Yield of specialized metabolites, including cannabinoids and terpenoids, strongly increased with increasing PPFD due to increases in both their concentrations and inflorescence yield. The inflorescence yield showed a proportional increase with PPFD, resulting in a constant light use efficiency. The higher inflorescence dry mass was mainly caused by an increase in total plant dry matter production, and to a lesser extent a larger fraction of dry mass partitioned to the inflorescences. Leaf photosynthesis was higher for plants grown at higher PPFD. This study shows that cannabis can use very high light intensity, resulting in high yield of specialized metabolites due to high inflorescence mass and metabolite concentrations.

1. Introduction

Indoor cultivation has been introduced to several horticultural crops, as it allows for control over plant quantity and quality through finely controlled growing conditions such as light, temperature, carbon dioxide, and air humidity (Park et al., 2022). Light in indoor cultivation can be designed according to many desired regimes, including light intensity, light spectrum, and photoperiod. Among others, medicinal cannabis (*Cannabis sativa* L.) as a high value crop, has benefited from indoor cultivation, as this allows for high quality and quantity control (Summers et al., 2021).

The primary focus of medicinal cannabis cultivation is to produce specialized metabolites, also named secondary metabolites, including

cannabinoids, terpenoids, and flavonoids. Groups of cannabinoids and terpenoids accumulate in the glandular trichomes, which are mainly present on mature female inflorescences (Andre et al., 2016; Livingston et al., 2020). The major bioactive forms of cannabinoids consist of delta-9-tetrahydrocannabinol (THC), cannabidiol (CBD), cannabigerol (CBG), cannabichromene (CBC), and cannabinol (CBN) (Richins et al., 2018), and there are more than 120 other cannabinoids which have been isolated from cannabis plants (ElSohly et al., 2017). Additionally, an abundance of terpenoids results in the unique aroma of cannabis (Hanuš and Hod, 2020). As these compounds are found together, they might have synergistic and/or entourage effects (Russo, 2011). From an ecological point of view, cannabis is hypothesised to produce cannabinoids and terpenoids to protect itself from the damage of UV light,

* Corresponding author.

E-mail address: leo.marcelis@wur.nl (L.F.M. Marcelis).

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pathogens, and pest insects (Gülck and Möller, 2020).

The production of specialized metabolites can be manipulated by influencing their concentrations and the yield of inflorescence. Biosynthesis of specialized metabolites is known to be light-dependent, with both light intensity and spectrum influencing their concentrations (Contreras-Avilés et al., 2024; Ouzounis et al., 2015; Ramakrishna and Ravishankar, 2011). In cannabis, however, studies on specialized metabolites have primarily focused on the effects of light spectrum rather than light intensity (Danziger and Bernstein, 2021; Magagnini et al., 2018; Morello et al., 2022; Rodriguez-Morrison et al., 2021a; Westmoreland et al., 2021). Light intensity can increase carbon gain via enhanced photosynthesis, and subsequently increases carbon-based specialized metabolites (Darko et al., 2014). It was reported that light intensity ranging from 120 to 1800 $\mu\text{mol m}^{-2} \text{s}^{-1}$ had no significant effect on concentrations of cannabinoids, but slightly increased those of total terpenes, in cannabis inflorescences (Rodriguez-Morrison et al., 2021b).

To promote the formation of inflorescences, cannabis requires a 'short-day' phase of maximum 12 hours of light per day during the generative phase (Zheng and Llewellyn, 2022). It has been recommended to use very high light intensities, with a photosynthetic photon flux density (PPFD) between 500 and 1500 $\mu\text{mol m}^{-2} \text{s}^{-1}$ PPFD (Chandra et al., 2017) to obtain a high daily light integral (DLI) which drives plant biomass production. These light intensities are much higher than those used for other horticultural crops: e.g., a PPFD of 250 $\mu\text{mol m}^{-2} \text{s}^{-1}$ is recommended for lettuce and basil for indoor cultivation (Pennisi et al., 2020) and 300 $\mu\text{mol m}^{-2} \text{s}^{-1}$ for dwarf tomato (Ke et al., 2023). A recent study using a light intensity gradient in a cannabis cultivation room found that increasing light intensity, ranging from 120 to 1800 $\mu\text{mol m}^{-2} \text{s}^{-1}$ PPFD using LED, resulted in higher inflorescence yield of cannabis (Rodriguez-Morrison et al., 2021b).

Photosynthesis uses most of the visible light absorbed by green plant tissue to power the assimilation of carbon dioxide into triose phosphates, which are then converted into carbohydrates (Calvin and Benson, 1948; Simkin et al., 2022). In many horticultural crops, there is a 0.7–1 % increase in harvestable biomass for every 1 % increase in light intensity (Marcelis et al., 2006). To maximize plant productivity, a light intensity above the photosynthetic light compensation point (i.e., the light intensity at which net photosynthesis rate is zero) and below the light saturation point of the photosynthetic response is often sought (Eichhorn Bilodeau et al., 2019; Simkin et al., 2019). The net photosynthesis rate of cannabis leaves increases with higher light intensities, and it remains unsaturated (i.e. it keeps increasing) even at levels up to 1500–2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ (Chandra et al., 2015, 2008). However, light intensity cannot be increased indefinitely; photosynthesis will eventually become light-saturated, caused by limitation of the photosynthetic machinery, leading to photodamage from excessive light (Raven, 2011; Zivcak et al., 2014).

The allocation of photoassimilates among plant organs determines harvestable yield and can be influenced by light intensity. In fruit crops, light intensity indirectly affects biomass partitioning by increasing the number of fruits or reducing rates of fruit abortion, thereby enhancing the overall sink strength of all fruits (Marcelis, 1996). In chrysanthemum, higher light intensity leads to a larger number of flowers, resulting in greater allocation of dry mass towards flowers (Heuvelink et al., 2002; Hosseinzadeh et al., 2021). In cannabis, higher light intensities have been shown to increase the harvest index, calculated as inflorescence yield over total aboveground biomass (Rodriguez-Morrison et al., 2021b), as well as the flower-to-leaf ratio (Potter and Duncombe, 2012).

Although it is known that an increase in light intensity results in increased inflorescence yield in cannabis, its impact on specialized metabolites remains unclear. Moreover, it is uncertain which components are responsible for these increases. This can be further investigated using a yield component analysis (Higashide and Heuvelink, 2009) to quantify the contributions of underlying components to yield.

Therefore, this study aims to quantify the effects of light intensity on the yield of specialized metabolites and to elucidate which plant traits explain these effects. We hypothesized that high light intensity increases specialized metabolites yield due to higher concentrations and higher inflorescence yield, where the increase of inflorescence yield is caused by higher total biomass production and higher dry mass partitioning towards inflorescences. We conducted an experiment in a climate-controlled chamber using light emitting diodes (LEDs) at three light intensities: 600, 800, and 1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ PPFD, applied during 8 weeks of the generative phase (i.e. short-day phase for flower induction and flower development) of medicinal cannabis.

2. Materials and methods

2.1. Plant cultivation

Medicinal *Cannabis sativa* L. cultivar 'Critical CBD' (chemotype II, with an intermediate THC/CBD ratio of approximately 0.5) plants were grown in a climate-controlled walk-in chamber. Plants were propagated by stem cuttings, which were rooted in stonewool plugs for 21 days (details on the rooting phase are described in Supplemental methods 1 (Sae-Tang et al., 2024)). Well-rooted cuttings were transplanted into 15 × 15 × 15 cm stonewool blocks (Grodan, Roermond, the Netherlands). The vegetative phase (i.e. long-day phase to promote vegetative growth) lasted from 0 to 11 days after transplanting (DAT); PPFD was 423 ± 24 $\mu\text{mol m}^{-2} \text{s}^{-1}$, with 13 % blue (400–500 nm), 6 % green (500–600 nm), 81 % red (600–700 nm), and <0.6 % far-red (700–800 nm) (Fig. S1), provided by a mixture of red-blue-white light-emitting diodes (LEDs, Green Power DRW MB 1.2, Philips, the Netherlands). The photoperiod was 18/6 hours (light/dark). The light (PPFD and spectrum) was measured at 45 cm above the table, which was at final canopy height during the vegetative phase. The PPFD was measured using a LI-250A light meter (LI-COR Inc., Lincoln, NE, USA) and the light spectrum was measured using a specbos 1210 spectrophotometer (JETI Technische Instrumente, Jena, Germany).

The planting density was kept constant at 9 plants m^{-2} throughout the whole experiment. Plants were pruned three times during cultivation. At 8 DAT, the apex was removed. At 11 DAT, the lowest side shoots were removed, such that the four upper side shoots were retained. Finally at 16 DAT, the lower second-order side shoots were removed to retain three second-order side shoots per side shoot.

During the vegetative phase (0–11 DAT), the climate was 28.7/26.3 °C (light/dark), 79/86 %RH, and CO₂ was 400 $\mu\text{mol mol}^{-1}$. During the generative phase (12–69 DAT), the climate was 28.0/26.7 °C, and 71/75 % RH, while CO₂ was gradually increased from 600 to 1200 $\mu\text{mol mol}^{-1}$, adding 200 $\mu\text{mol mol}^{-1}$ every 2 weeks. Realised air temperature and RH per light treatment and week are shown in Table S1. The maximum differences of average realised air temperature and RH between treatments were 1.5/0.3 °C and 5.3/1.2 %RH (light/dark values). Before transplanting, stonewool blocks were pre-soaked in a nutrient solution with an EC of 1.6 dS m^{-1} and a pH of 5.8. The solution consisted of the following macro- and micronutrients: 1.25 mM NH₄⁺; 6.2 mM K⁺; 1.9 mM Ca²⁺; 0.9 mM Mg²⁺; 10.5 mM NO₃⁻; 0.85 mM SO₄²⁻; 0.85 mM PO₄³⁻; 60 μM Fe²⁺; 20 μM Mn²⁺; 3 μM Zn²⁺; 20 μM B³⁺; 0.5 μM Cu²⁺; 0.5 μM Mo²⁺. Irrigation from transplanting onward was supplied with the same nutrient solution by discharge-regulated drippers into the stonewool blocks, each discharge was 100 mL. Frequency of dripping ranged from four to six cycles per day, and it was adjusted to ensure that all plants received sufficient water and nutrients. The EC value was first raised to 2 dS m^{-1} (0–11 DAT), then further raised to 2.5 dS m^{-1} during the generative phase (12–69 DAT), by increasing the concentration of macronutrients, while keeping the ratios between individual macronutrients constant.

2.2. Light intensity treatments

During the 8 weeks of the generative phase (12–69 DAT), three light intensity treatments were applied: 600, 800, or 1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ PPFD. Realized light intensities were 595 ± 11 , 791 ± 10 , and 991 ± 15 $\mu\text{mol m}^{-2} \text{s}^{-1}$ PPFD respectively, measured at 75 cm above the table, which was at final canopy height. The light was provided by the same fixtures that were used in the vegetative phase; the light spectrum was therefore similar (described above). The photoperiod was 12/12 hours (light/dark). These three light treatments were randomly assigned to each compartment within the climate room and were partitioned by white plastic screens. Each compartment measured 3.3 m² (replicate 1) and 2 m² (replicate 2).

2.3. Growth and morphology measurements

Final harvest was conducted after 8 weeks into the generative phase (67–69 DAT). Plant height was measured as the longest distance from the base of the stem towards the top of the canopy. The diameter of the main stem was determined at 1 cm above the top of the stonewool block. Each plant was dissected into leaves, stems, and inflorescences. Inflorescence leaves and stems were trimmed by an electronic trimmer (Bowl Trimmer 40 cm, Dutchmasters Fertraso, the Netherlands), and their trimmed mass was added to that of the stems. Leaf area per plant was determined using a LI-3100 C area meter (LI-COR Inc., Lincoln, NE, USA). Inflorescence mass was determined after trimming, measured as fresh mass, approximately 10 % moisture dry mass (air-dried, as is the commercial standard), and dry mass. Drying to 10 % moisture of inflorescences took place for five days in a dark climate-controlled room with ample ventilation, at 45 % RH, and 25 °C. Dry mass was determined after drying the material at 70 °C for 24 hours, followed by 105 °C for 72 hours in a ventilated oven. Dry matter content was calculated as the ratio between dry and fresh mass. Light use efficiency (LUE) was calculated as dry mass accumulated over the whole cultivation cycle divided by cumulative incident mol of PPFD during the generative phase (treatment period). Plant compactness was calculated as the ratio of total plant dry mass to plant height. Specific leaf area (SLA; cm² g⁻¹) was calculated as the ratio between leaf area and leaf dry mass.

2.4. Photosynthesis light response curves

Photosynthesis light response curves were determined at 3, 5, and 7 weeks into the generative phase in replicate 2, using a LI-6800 Portable Photosynthesis System (LI-COR) equipped with a Multiphase Flash Fluorometer and Chamber (LI-COR part No. 6800-01 A, area 2 cm²). Measurements were taken on one leaf per plant using the youngest, fully developed, unshaded leaves, with 3–5 plants chosen at random per light treatment (n=3–5). At 3 weeks into the generative phase, net photosynthesis rate (*A*) ($\mu\text{mol CO}_2 \text{m}^{-2} \text{s}^{-1}$) was measured at 2000, 1500, 1000, 800, 600, 400, 200, 100, and 0 $\mu\text{mol m}^{-2} \text{s}^{-1}$, with each step taking 3–5 min. At 5 and 7 weeks into the generative phase, a light intensity step at 3000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ was added. Light was provided by a mixture of red (87 %) and blue (13 %) LEDs in the fluorometer (at the light step of 3000 $\mu\text{mol m}^{-2} \text{s}^{-1}$, the ratio changed to 90 % red and 10 % blue, as the capacity of blue LEDs was exceeded at that intensity). Leaves were first adapted to the highest light intensity (either 2000 or 3000 $\mu\text{mol m}^{-2} \text{s}^{-1}$) for 15 min, or until *A* and stomatal conductance (*g_s*) were stable. Data were logged after stability was reached, i.e. when the slope of ΔCO_2 was <0.1 over a period of 20 s. A rolling average of 10–15 s was used. CO₂ concentration was 800 $\mu\text{mol mol}^{-1}$, air flow rate was 500 $\mu\text{mol s}^{-1}$, RH was 60–70 %, and air temperature was 27 °C, yielding a leaf temperature of 27–30 °C across all light steps. Light response curves were fitted to a non-rectangular hyperbola (Ogren and Evans, 1993), yielding the parameters maximum net photosynthesis rate (*A_{max}*; $\mu\text{mol CO}_2 \text{m}^{-2} \text{s}^{-1}$), (α ; $\mu\text{mol CO}_2 \mu\text{mol photon}^{-1} \text{s}^{-1}$), a curvature parameter (θ), and dark respiration (*R_d*; $\mu\text{mol CO}_2 \text{m}^{-2} \text{s}^{-1}$) (Eq. 1). The light

compensation point (*LCP*; $\mu\text{mol m}^{-2} \text{s}^{-1}$ PPFD) was determined from *R_d* and α .

$$A = -R_d + \frac{\alpha \bullet \text{PPFD} + A_{\text{max}} - \sqrt{(\alpha \bullet \text{PPFD} + A_{\text{max}})^2 - 4\theta \text{PPFD} \bullet A_{\text{max}} \bullet \alpha}}{2\theta} \quad (1)$$

2.5. Specialized metabolites

Fresh inflorescences were collected from the top 5 cm of inflorescences in 6, 7, and 8 weeks into the generative phase, and stored at –80 °C. A total of five biological samples per treatment was prepared, in which each sample was pooled from 4 to 6 plants. Per sample, 200 mg fresh inflorescences were homogenized before extraction in 2 mL of *n*-Hexane with 0.2 g L⁻¹ of squalene as an internal standard (IS). Sample extracts were sonicated for 10 min in an ultrasonic bath (Branson 2800; Sigma-Aldrich, St. Louis, MI, USA), subsequently filtered through a column containing siloxilated glass-wool and anhydrous sodium sulphate in a Pasteur pipet and collected in a 2 mL glass vial. Two μL of each filtered extract was analysed in splitless mode on a Gas Chromatography-Mass Spectrometry (GC-MS) Agilent (7890) equipped with a 30-m length $\times$ 0.25-mm inner diameter, 0.25- μm film thickness column (Zebtron, 5MS) and a mass-selective detector (model 5972 A, Hewlett-Packard). The GC was programmed at an initial temperature of 60 °C for 2 min, with a ramp of 5 °C min⁻¹ to 250 °C, and then with 10 °C min⁻¹ to 280 °C and final time of 5 min. The injection port, interface, and MS source temperatures were 250 °C, 290 °C, and 180 °C, respectively, and the Helium inlet pressure was controlled with an electronic pressure control to achieve a constant column flow of 1 mL min⁻¹. The ionization potential was set at 70 eV, and scanning was performed from 45 to 400 at. mass unit (amu). Metabolites were identified by comparing their mass spectra with those of the National Institute of Standards and Technology mass spectral library (NIST) MS search 2.0 and an in-house spectral library generated with authentic standards. In addition, the relative retention times of individual metabolites were compared to those of Adams (2017). Detected compounds were classified as either terpenoids or cannabinoids. Relative abundances (IS ratio response) of individual compounds were determined by the area under the curve, and normalized by IS and fresh mass multiplied by the ratio between dry to fresh mass (as described in Section 2.3). Total terpenoids were calculated as the sum of metabolites at retention time (RT) 6.2–26.2 min, from which the proportion of total monoterpenoids at RT 6.2–11.3 min and total sesquiterpenoids at RT 19.5–26.2 min was determined by dividing to the total terpenoid abundance. Total cannabinoids were a sum of metabolites at RT 35.0–43.2 min.

2.6. Yield component analysis

The effect of light intensity on specialized metabolites yield was analysed by dissecting the effects on the underlying components, as illustrated in Fig. 1. The yield of specialized metabolites per plant is defined by the product of inflorescence dry mass and the concentrations of specialized metabolites. The inflorescence dry mass is the product of total plant dry mass and fraction of dry mass partitioned to the inflorescences. The total plant dry mass is the result of fraction intercepted light (depends on leaf area index) and net photosynthesis rate, measured at its growth light intensity treatments (e.g., 600, 800, 1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$) at 5 weeks into the generative phase. These values were extracted from the light response curves of each treatment. The leaf area index is the product of leaf dry mass per m² ground area and specific leaf area. A linear relationship was fitted to yield and each of the yield components with PPFD as regressor, and this regression was utilized to calculate the percentage change of each component from 600 to 1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ PPFD.

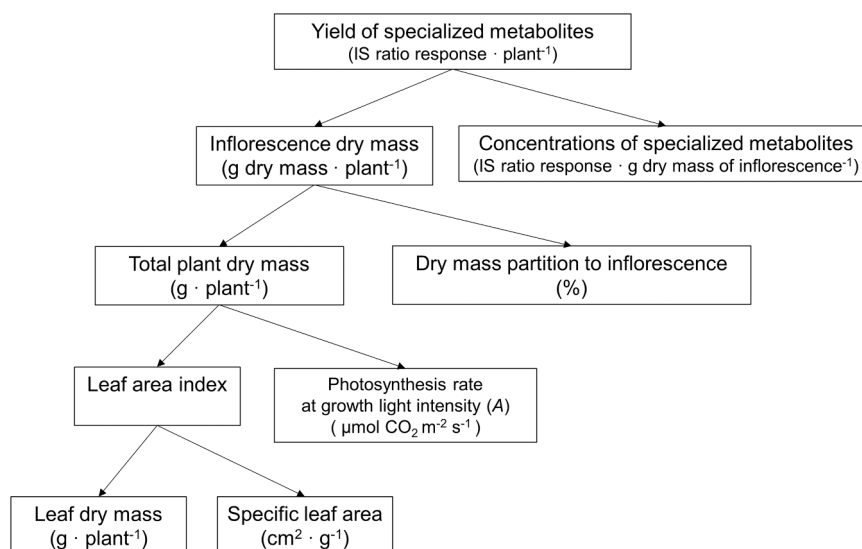


Fig. 1. General scheme of a top-down analysis of specialized metabolites yield, separated into underlying components. A represents net photosynthesis rate measured at growth light intensity (i.e. 600, 800, and 1000 $\mu\text{mol m}^{-2}\text{s}^{-1}$), 5 weeks into the generative phase. Units are provided between brackets. Metabolites were calculated as internal standard (IS) ratio response.

2.7. Statistical set-up and analysis

The experiment was performed twice in the same climate chamber, with a shift in time. The replicate experiments represented two blocks, each having a new randomization. There were 30 and 18 replicate plants per experiment (i.e., per treatment per experiment; hence 60 and 36 plants per treatment for the two experiments together) in replicate experiment 1 and 2, respectively. For morphological traits and dry mass, a one-way Analysis of Variance (ANOVA) in two blocks was conducted using the average values for each experimental unit. The effect of light intensity was partitioned into a linear and a quadratic component (orthogonal polynomial contrasts). For the concentrations of specialized metabolites, average values of 5 biological samples per experimental unit were used in a two-way ANOVA in two blocks, with sampling week and light intensity (partitioned into a linear and a quadratic component) as factors (no interaction). For the photosynthesis parameters, which were determined only in replicate 2, the measurement on each plant ($n=3-5$) was considered as an independent replicate in a one-way ANOVA. Statistical tests were conducted at $P=0.05$. The normality of the residuals was examined with Shapiro-Wilk test at $P=0.05$. For the photosynthesis parameters, homogeneity of residuals was tested with Bartlett's test at $P=0.05$, while homogeneity for the other traits was assumed as it could not be tested because of the small number of replicates ($n=2$). These assumptions were met in all cases. All tests were performed using Genstat 21st edition (VSN International, Hemel Hempstead, UK). Multivariate analysis (MetaboAnalyst 5.0 online platform; (Chong et al., 2019)) of relative abundance of specialized metabolites was performed using individual biological replicates from both repetitions ($n=10$). These metabolites data were analysed after log transformation and mean centring.

3. Results

3.1. Concentrations of specialized metabolites in inflorescences

In total, 56 specialized metabolites that were either terpenoids or cannabinoids were detected in inflorescences. Unsupervised principal component analysis (PCA) did not show a separation of specialized metabolite profiles between the three PPFD treatments (Fig. S2). However, using a sparse partial least squares-discriminant analysis (SPLS-DA) as a supervised model showed that the groups of 600 and 1000 $\mu\text{mol m}^{-2}\text{s}^{-1}$ PPFD could be distinguished from each other, while results from the intermediate PPFD (800 $\mu\text{mol m}^{-2}\text{s}^{-1}$ PPFD) were to a relatively large extent overlapping with those of the other two PPFD levels (Fig. 2A-C). This separation between PPFD treatments was more apparent in the early phase, after 6 weeks into the generative phase (Fig. 2A). Variability in projection scores of individual metabolites, for example of Component 1, indicated that multiple metabolites were responsive to increasing PPFD, and this was most obvious for sesquiterpenoids (Fig. 2D-F). Analyzing metabolite abundance with ANOVA, considering sampling weeks and experimental blocks, showed a linear increase in total terpenoids, total cannabinoids, and also multiple individual compounds with increasing PPFD (Fig. 3). The observed responses showed similar trends in both replicate experiments (Fig. S3-S4). Considering the individual weeks, a significant increase of CBN, CBG, and some terpenoids, including β -myrcene, β -caryophyllene, α -humulene, β -farnesene, and γ -eudesmol was found after 6 weeks into the generative phase, compared to their respective abundances after 7 or 8 weeks (Fig. S5).

3.2. Plant dry mass, partitioning, and morphology

The total dry mass of the aboveground plant, as well as the individual dry masses of the inflorescences, leaves, and stems, all showed linear increases with increasing PPFD (Fig. 4A). The fraction of dry mass partitioned to the inflorescences also increased linearly with PPFD, though this effect was small (only 3.4 % increase in partitioning when PPFD increased from 600 to 1000 $\mu\text{mol m}^{-2}\text{s}^{-1}$). No significant change was found in the partitioning to leaves nor to stems (Fig. 4B). Dry matter content of neither whole plant nor individual organs was affected by light intensity (Fig. 4C). Also, remarkably the light use efficiency (LUE) of both inflorescences and whole plants remained constant over the PPFD range from 600 to 1000 $\mu\text{mol m}^{-2}\text{s}^{-1}$ (Fig. 4D, Fig. S7). Plant height was unaffected (Fig. 5A, Fig. S8), while the diameter of the main stem increased linearly with PPFD (Fig. 5B). High PPFD increased plant compactness (Fig. 5C). Specific leaf area showed a linear decrease with rising PPFD, while leaf area per plant remained stable (Fig. 5D). The responses of plant biomass and morphology to light intensity were similar in both replicate experiments (Fig. S9-S12).

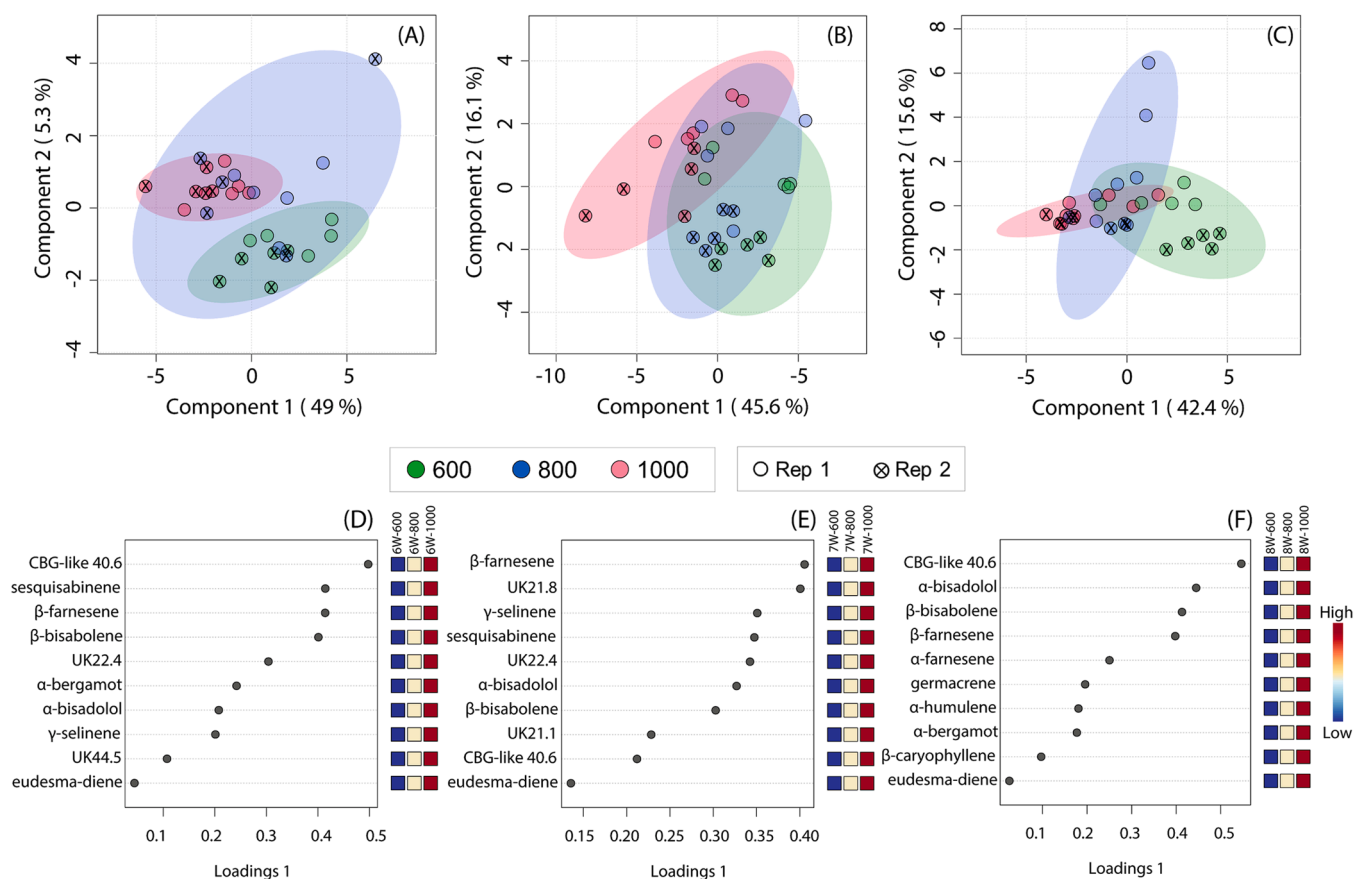


Fig. 2. Sparse partial least-squares discriminant analysis (sPLS-DA) plots (A-C) and corresponding variable in projection plots indicating the most discriminating compounds of component 1 (D-F) of specialized metabolites in inflorescences of medicinal cannabis grown under 600 (green points), 800 (blue points), and 1000 (red points) $\mu\text{mol m}^{-2} \text{s}^{-1}$ PPFD at 6 (A, D), 7 (B, E), and 8 weeks (C, F) into the generative phase. The data are based on variation in the relative abundance of 56 specialized metabolites detected by GC-MS from 5 biological samples per light treatment per replicate experiment, 2 replicates were analyzed together ($n=10$). The circles represent replicate experiment 1, while the circles with cross represents replicate experiment 2 (A-C). When the compounds cannot be matched with those of the NIST library, they are represented as unknown (UK) with the retention time. The color coding in D-F indicates the relative abundance from lowest (blue) to highest (red).

3.3. Photosynthesis light response curves

Increasing PPFD increased photosynthetic capacity, but this effect depended on plant developmental stage: At 5 weeks into the generative phase, net photosynthesis rate (A) in the range of 1000–3000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ was significantly higher in the 800–1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ growth PPFD treatments compared to the 600 $\mu\text{mol m}^{-2} \text{s}^{-1}$ (Fig. 6B). Maximum photosynthesis rate (A_{max}) increased linearly with growth light PPFD at 3 and 5 weeks, but not at 7 weeks (Fig. 6D). Dark respiration (R_d) and the light compensation point (LCP) increased linearly with PPFD of growth light at 3 weeks (Fig. 6F). The apparent quantum yield (α) was not affected by PPFD of growth light (Fig. 6E). After 7 weeks, there was no effect of PPFD on any of the parameters of the light response curve (Fig. 6C-G). Remarkably, A was not saturated (i.e. kept increasing with increasing PPFD even at the highest measurement PPFD (2000–3000 $\mu\text{mol m}^{-2} \text{s}^{-1}$), regardless of treatment and measurement week (Fig. 6A-C).

3.4. Yield component analysis

When PPFD increased from 600 to 1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$, metabolites yield per plant went up by 140 % for cannabinoids and 214 % for terpenoids, due to increases in both inflorescence yield and concentrations of metabolites in the inflorescences. Remarkably, increasing the light intensity during the generative phase by 67 % led to a similar increase (69 %) in dry mass of the inflorescences (Fig. 7). This increased

inflorescence dry mass was caused by an increase in total plant dry mass rather than a higher fraction of dry matter partitioned to the inflorescences. This increase in total plant dry mass was primarily attributed to an increase in photosynthesis rate, measured at growth light intensities, and was also partially attributed to a rise in leaf area. The higher leaf area was related to an increase in leaf dry mass, while specific leaf area (SLA) decreased.

4. Discussion

4.1. Concentrations of specialized metabolites increased strongly with increasing PPFD

The yield of specialized metabolites - both terpenoids and cannabinoids - in fresh inflorescences increased strongly with increasing PPFD, due to higher inflorescence yield as well as higher concentrations of these metabolites. The increased concentrations of specialized metabolites in response to a higher light intensity was found in several other crops (Bian et al., 2015). An increase in light intensity increases available photoassimilates, which are not only essential for plant growth, but also serve as precursors for specialized metabolites biosynthesis (Darko et al., 2014) as well as energy source for their biosynthesis (Zavala and Ravetta, 2001). Moreover, elevated light intensity enhances the necessity for photoprotection (Bassi and Dall'Osto, 2021) to prevent oxidative stress by promoting specialized metabolite biosynthesis (Ramakrishna and Ravishankar, 2011). Concentrations of both sesquiterpenoids and

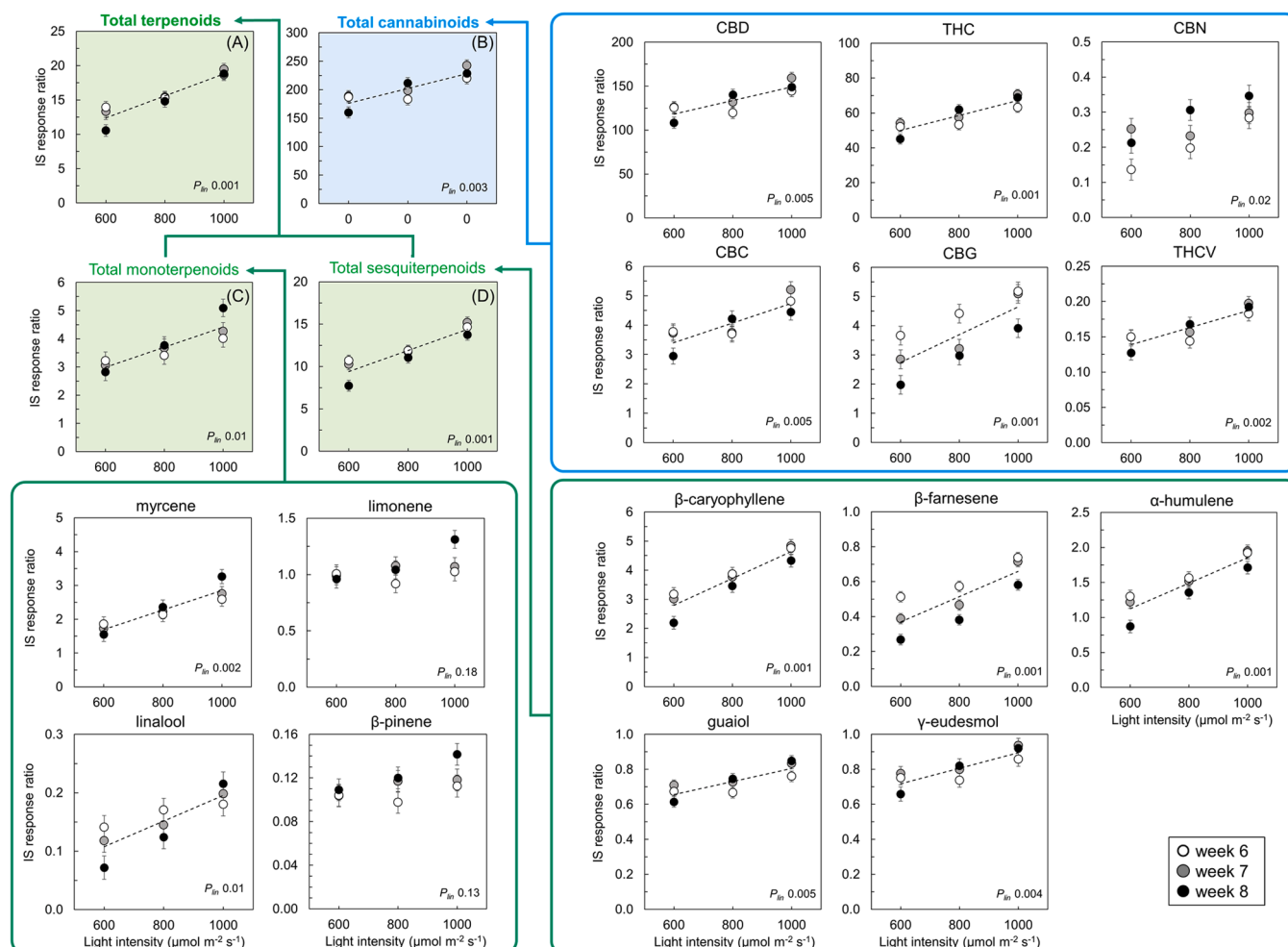


Fig. 3. Relative abundance of specialized metabolites in inflorescences of medicinal cannabis grown under 600, 800, and 1000 $\mu\text{mol m}^{-2}\text{s}^{-1}$ PPFD after 6 (white points), 7 (grey points), and 8 (black points) weeks into the generative phase. The metabolites are calculated to dry mass basis. Two major groups are total terpenoids (A) and total cannabinoids (B). Total terpenoids consist of two subgroups: total monoterpenoids (C), including myrcene, limonene, linalool, and β -pinene, and total sesquiterpenoids (D), including β -caryophyllene, β -farnesene, α -humulene, guaial, and γ -eudesmol. Total cannabinoids consist of CBD, THC, CBN, CBC, CBG, and THCV. Data are averages of on two replicate experiments ($n=2$), each consisting of 5 pooled biological samples per treatment. Data of each replicate experiment separately are shown in Fig. S3 and S4. Error bars represent standard error of means (SEM) based on the common variance and are visible when larger than the symbols. When linear effects of light intensity for the overall 3 weeks were significant at $P=0.05$, trendlines are depicted. The abundance of these metabolites on fresh mass basis shown in Fig. S6.

monoterpenoids increased linearly with PPFD in plants during the 6–8 weeks into the generative phase (Fig. 3). Sesquiterpene biosynthesis was found to be positively correlated with photosynthetic efficiency and the production of stress-related phytohormones in *Atractylodes lancea* (Guo et al., 2022). In our study, total terpenoids concentration in fresh inflorescences increased by 76 % with a 67 % increase in light intensity. Rodriguez-Morrison et al. (2021b) observed a smaller effect of light intensity on dry inflorescences, with only a 43 % increase when light intensity was increased by 1400 %. This increase was mainly observed for individual monoterpenoids like limonene and myrcene. However, during drying of inflorescences, a considerable loss of (mono-) terpenoids can occur, which may have reduced the effects reported by (Rodriguez-Morrison et al., 2021b). A higher light intensity led to increased concentration and yield of total essential oil in leaves of mint (*Mentha arvensis* L.) that were mainly contributed of monoterpenoids (de Souza et al., 2015). In addition, high light intensity increased concentration of a diterpenoid steviol glycoside in leaves of *Stevia rebaudiana*, which was linked to an up-regulation of related biosynthesis genes (Hernández et al., 2022). We found concentrations of detected cannabinoids increased with PPFD. In contrast, other studies found that light intensity had no effect on THC (Rodriguez-Morrison et al., 2021b;

Vanhove et al., 2011). In some cases, cannabinoid concentration decreased due to a dilution effect caused by increased inflorescence yield (Bevan et al., 2021), but this was not observed in our study. Such contrasting effects of increased light intensity between studies may partly be explained by the cellular localization of different metabolites. Sesquiterpenoids are mostly synthesized in the cytosol from the precursor farnesyl pyrophosphate (FPP), which is generated in the mevalonate (MVA) pathway. On the other hand, monoterpenoids and cannabigerol (CBG) are synthesized in the plastids from geranyl pyrophosphate (GPP), which is generated in the methyl-D-erythritol phosphate (MEP) pathway (Romero et al., 2020), and the latter is suggested to be light-induced (Tholl, 2015). The transformation of CBG to other cannabinoids occurs in the secretory vesicles (Romero et al., 2020). Furthermore, the variation in the increase of different metabolites in response to light intensity might be related to a distinct effect of light on an induction of different trichome types. For instance, sessile glandular trichomes tend to accumulate more sesquiterpenoids, while stalked trichome types accumulate more monoterpenoids (Livingston et al., 2020). As genes involved in both terpenoid and cannabinoid biosynthesis were identified in a co-expression network (Zager et al., 2019), a light effect on genes related to the MEP and/or the MVA pathway might

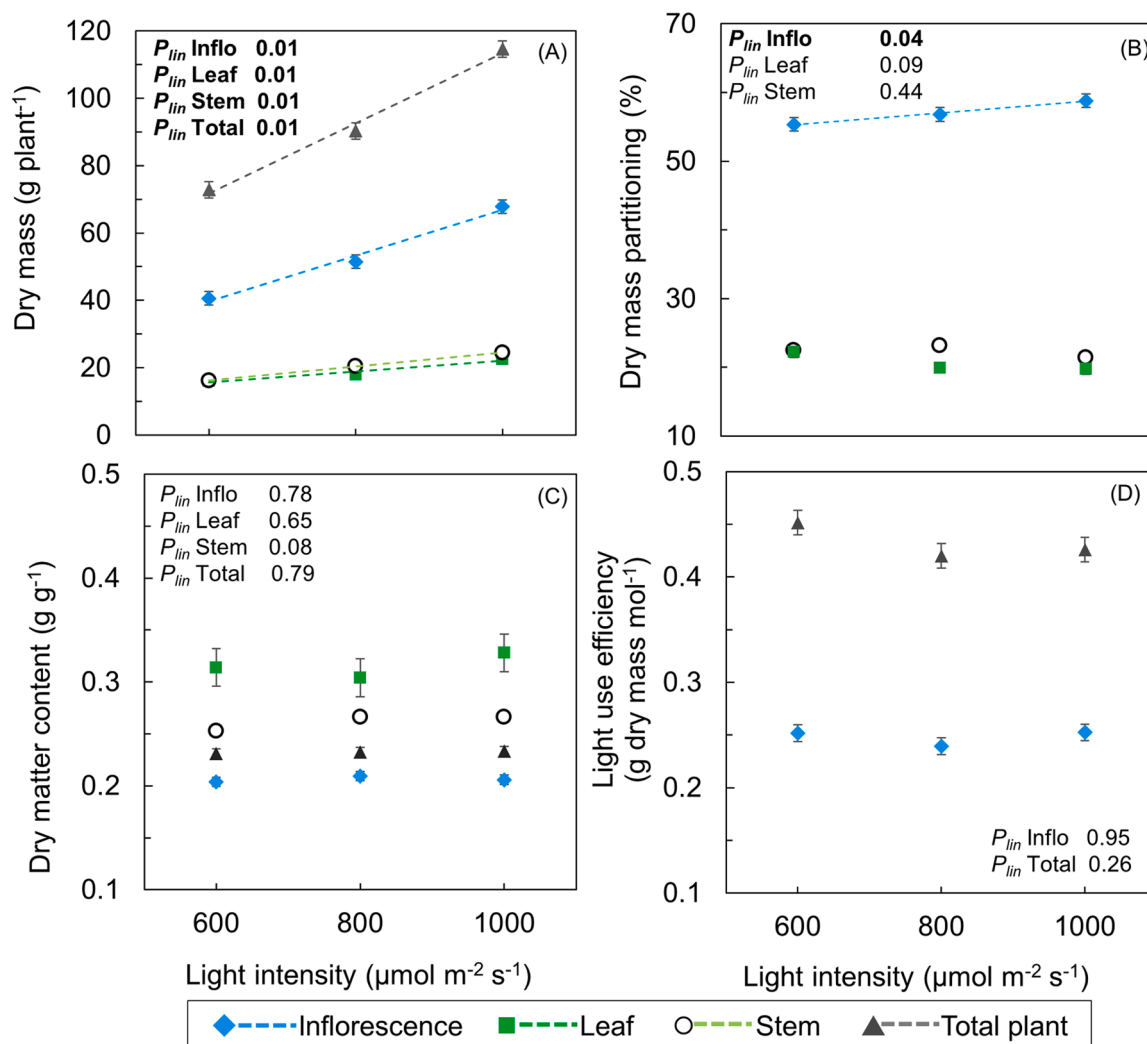


Fig. 4. Plant dry mass (A), dry mass partitioning (B), dry matter content (C), and light use efficiency (D) of medicinal cannabis grown at 600, 800, and 1000 $\mu\text{mol m}^{-2}\text{s}^{-1}$ PPFD after 8 weeks into the generative phase. Data based on two replicate experiments ($n=2$), each consisting of 30 (replicate 1) and 18 (replicate 2) plants per treatment. Data of each replicate experiment separately are shown in Fig. S9 and S10. Error bars represent standard error of means (SEM) based on the common variance and are visible when larger than the symbols. When linear effects of light intensity were significant at $P=0.05$, trendlines are depicted.

affect the mutual balance between terpenoids and cannabinoids. To what extent these factors determine the responses in specialized metabolite abundances as found in our study needs further exploration of the regulation of biosynthetic genes in response to light intensity.

4.2. Inflorescence dry mass increased proportionally with PPFD, while LUE remained unchanged

Besides the higher concentrations of metabolites, a higher inflorescence yield contributed significantly to the increase in the yield of metabolites. This higher inflorescence yield at higher PPFD aligns with other studies on cannabis (Eaves et al., 2020; Potter and Duncombe, 2012; Rodriguez-Morrison et al., 2021b; Vanhove et al., 2011). In our experiment, a 1 % increase in PPFD resulted in a ~1 % increase in inflorescence yield (Fig. 7), which matches with the 1 %-rule of thumb in harvestable yield used and observed for several greenhouse crops (Marcelis et al., 2006). This also implies that the light use efficiency (LUE) remained constant at about 0.25 g inflorescence dry mass per mol of light (g mol^{-1}) within a PPFD ranging from 600 to 1000 $\mu\text{mol m}^{-2}\text{s}^{-1}$. This constant LUE at PPFD levels up to at least 1000 $\mu\text{mol m}^{-2}\text{s}^{-1}$ is remarkable, as it is rare at such high light intensities. This stability in LUE may partially be attributed to optimal growth conditions that were

likely achieved in our setup, including factors such as CO_2 , water, temperature, and nutrients (Jin et al., 2023; Medlyn, 1996), but additionally testifies to the remarkable ability of cannabis to use high light intensities productively (see below). In many other species, LUE usually decreases with increasing PPFD even at lower PPFD ranges (i.e. for 200 up to 750 $\mu\text{mol m}^{-2}\text{s}^{-1}$ in lettuce; 250 up to 300 $\mu\text{mol m}^{-2}\text{s}^{-1}$ in basil (Carotti et al., 2021; Heuvelink et al., 2002; Jeong et al., 2003; Pennisi et al., 2020)). This response is likely due to a saturation of leaf photosynthesis response to light intensity (Fu et al., 2012).

4.3. Higher inflorescence dry mass was mainly due to an increase in total plant dry matter production, with a small contribution of increased harvest index

The increase in inflorescence dry mass with increasing PPFD was mainly due to increased total plant dry matter production, while there was a small contribution from a higher dry mass partitioning to inflorescences (i.e. higher harvest index) (Fig. 7). The increase in total plant dry mass correlated with an increase in leaf photosynthesis. Specific leaf area (SLA) decreased with increasing PPFD, indicating thicker leaves, a common crop response to high light intensity (Evans and Poorter, 2001; Poorter et al., 2019). This leaf thickening, caused by

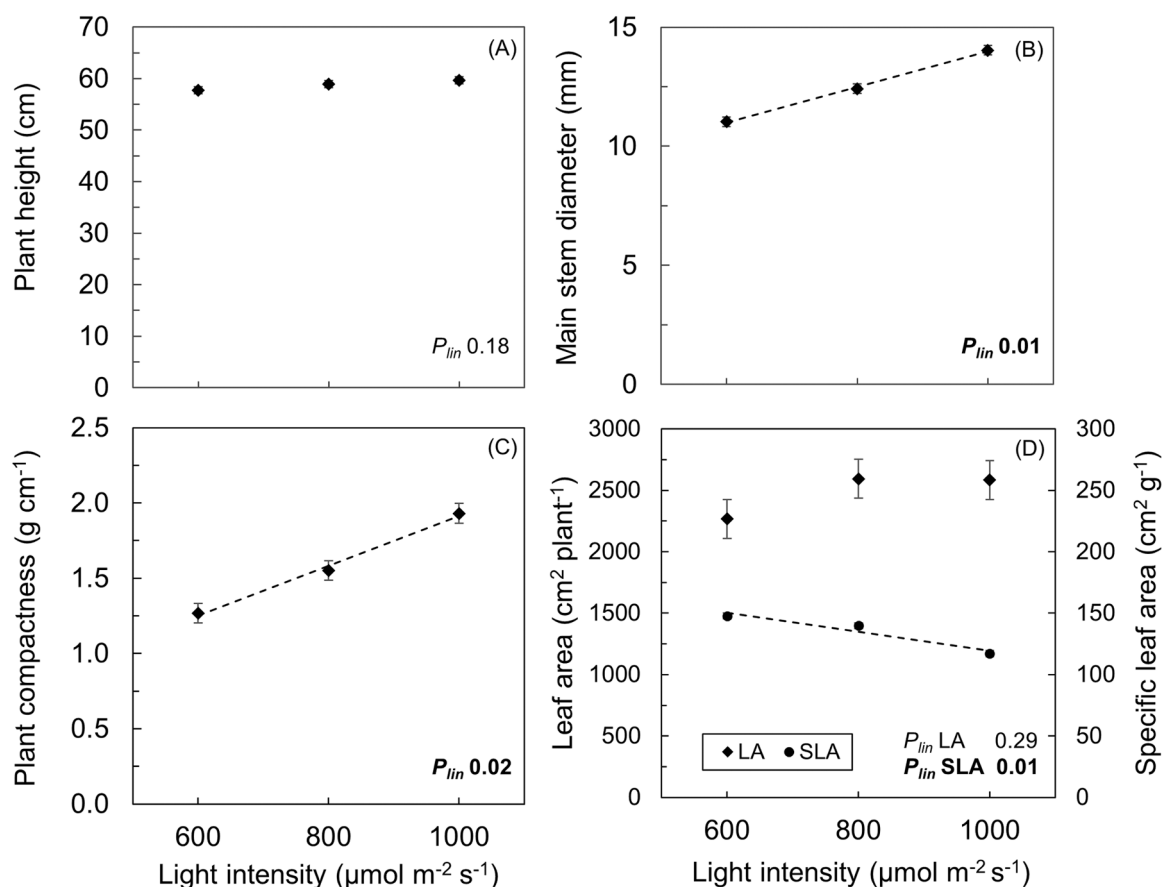


Fig. 5. Plant height (A), diameter of the main stem (B), plant compactness (C), leaf area (LA) and specific leaf area (SLA) (D) of medicinal cannabis grown at 600, 800, and 1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ PPFD after 8 weeks into the generative phase. Data are based on 2 replicate experiments ($n=2$), each consisting of 30 (replicate 1) and 18 (replicate 2) plants per treatment. Data of each replicate experiment separately are shown in Fig. S11 and S12. Error bars represent standard error of means (SEM) based on the common variance and are visible when larger than the symbols. When linear effects of light intensity were significant at $P=0.05$, trendlines are depicted.

larger palisade cells and more spongy parenchyma layers, may improve photosynthetic capacity via an increase in the number of chloroplasts and photosynthetic proteins per chloroplast (Murchie et al., 2022; Zheng and Van Labeke, 2018). PPFD did not influence plant height, although total plant dry mass increased, hence higher PPFD promoted plant compactness. Simultaneously, the diameter of the main stem increased with PPFD, which may be a result of supporting the higher inflorescence weight. Flowering triggers secondary bast fiber formation and an intense lignification of the fiber to prepare the plant to hold heavier inflorescences (Westerhuis et al., 2019). Although the measured air temperature and relative humidity varied slightly among PPFD treatments (Table S1; 24-h average temperature differed less than 1°C), we anticipated little interference with the positive effect of increasing PPFD on plant growth and yield. This study was performed with a spectrum consisting of 13 % blue, 6 % green, 81 % red, and 0.6 % far-red. However, Holweg et al. (2024) showed that effects of PPFD on plant dry matter production and inflorescence yield can interact with light spectrum.

4.4. Cannabis is a plant with exceptionally high photosynthetic capacity

We found that cannabis is a high photosynthesis capacity species, as A did not saturate even at 3000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ PPFD and showed values of 40–55 $\mu\text{mol CO}_2 \text{m}^{-2} \text{s}^{-1}$ (Fig. 6A–C). Even though light response curves were conducted at high CO_2 (800 $\mu\text{mol CO}_2 \text{mol}^{-1}$), these rates are higher than reported for many other C_3 plants (<40 $\mu\text{mol CO}_2 \text{m}^{-2} \text{s}^{-1}$ for tomato (Pan et al., 2020); <30 $\mu\text{mol CO}_2 \text{m}^{-2} \text{s}^{-1}$ for strawberry (Li et al., 2020), measured at similar light and CO_2 condition (2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ PPFD and 800 $\mu\text{mol CO}_2 \text{mol}^{-1}$)).

An example of a C_3 crop with high photosynthetic capacity is sunflower (English et al., 1979), which was found to exhibit A of $\sim 45 \mu\text{mol CO}_2 \text{m}^{-2} \text{s}^{-1}$ (Atkins and Boldt, 2022). Garassino et al. (2022) and Taylor et al. (2023) proposed *Hirschfeldia incana* as a model plant of C_3 species with high photosynthetic LUE at high irradiance, with an A of 50–52 $\mu\text{mol CO}_2 \text{m}^{-2} \text{s}^{-1}$, measured at 2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ PPFD and 400 $\mu\text{mol CO}_2 \text{mol}^{-1}$. To our knowledge, these rates are only surpassed by *Amaranthus palmeri*, a C_4 desert annual, whose A was shown to reach $\sim 80 \mu\text{mol CO}_2 \text{m}^{-2} \text{s}^{-1}$ at 2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ PPFD and 325 $\mu\text{mol CO}_2 \text{mol}^{-1}$ (Ehleringer, 1983). Altogether, we posit that cannabis is likely a photosynthesis extremophile, i.e. a species that can productively use very high light intensities for growth.

In addition, cannabis plants grown at higher PPFD showed a higher leaf photosynthetic capacity, in line with what is often found in other crops (Poorter et al., 2019). However, unlike many other crops that tend to show saturation of A well below full sunlight (2000 $\mu\text{mol m}^{-2} \text{s}^{-1}$; (Taylor et al., 2023), A in cannabis leaves did not saturate at even 3000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ (Fig. 6B). The fact that biomass accumulation of both inflorescences and total plant dry mass was unsaturated at the highest PPFD of 1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$, indicates that cannabis is a light-demanding species that can grow under extremely high light intensity without yield reduction. Rodríguez-Morrison et al. (2021b) found that A of cannabis leaves started to saturate at 1000 $\mu\text{mol m}^{-2} \text{s}^{-1}$ PPFD of measurement light. However, their measurement was done at 400 $\mu\text{mol CO}_2 \text{mol}^{-1}$, which is only half of that in our experiment (i.e. at 800 $\mu\text{mol CO}_2 \text{mol}^{-1}$, commonly used in commercial practice (Zheng and Llewellyn, 2022)). This suggests that increasing light intensity in

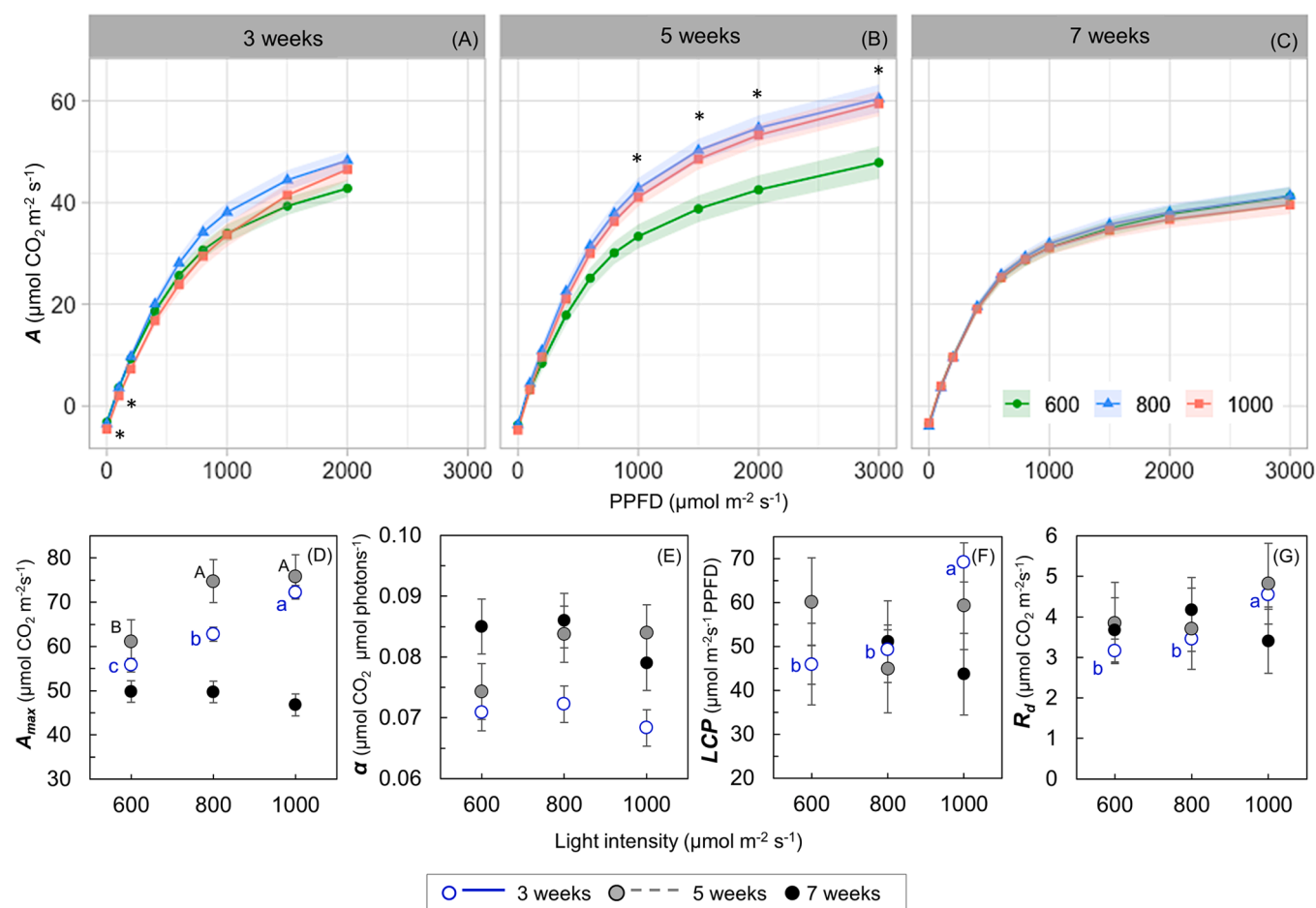


Fig. 6. Light response curves of net photosynthesis rate (A) for medicinal cannabis grown under 600, 800, and 1000 $\mu\text{mol m}^{-2} \text{ s}^{-1}$ PPFD at 3 (A), 5 (B), and 7 (C) weeks into the generative phase. Parameters maximum estimated photosynthesis rate (A_{max}) (D), apparent quantum yield (α) (E), light compensation point (LCP) (F), and dark respiration (R_d) (G) were derived from light response curves. Symbols represent means of 3–5 plants measured in replicate experiment 2. Ribbons in A–C and error bars in D–G represent standard error of means (SEM) ($n=3-5$). When effects of light intensity were significant at $P=0.05$, asterisks (A–C) or different letters (D–G) are depicted.

commercial cannabis production should be combined with sufficient supplemental CO_2 in order to increase inflorescence dry mass. Dark respiration was about 3–5 $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$, which was relatively high compared to other crops (at similar temperature), for example, 1–2 $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$ has been reported for tomato (Kaiser et al., 2019) and 1–3 $\mu\text{mol CO}_2 \text{ m}^{-2} \text{ s}^{-1}$ for chrysanthemum (Zhou et al., 2013). This high respiration rate in cannabis can be expected due to its high photosynthetic capacity.

5. Conclusions

Yield of specialized metabolites, including cannabinoids and terpenoids, strongly increased with increasing light intensity (600–1000 $\mu\text{mol m}^{-2} \text{ s}^{-1}$) resulting from increases in both their concentrations and inflorescence yield. Within this range of light intensities, the inflorescence yield showed a proportional increase with light intensity, meaning that light use efficiency remained constant. Higher inflorescence yield was mainly due to higher total plant dry matter production, and to a lesser extent due to an increase in the fraction of dry mass partitioned to the inflorescences. Cannabis is a species with exceptionally high photosynthetic capacity, in which leaf photosynthesis rate kept on increasing up to the highest measuring light level (3000 $\mu\text{mol m}^{-2} \text{ s}^{-1}$).

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CRediT authorship contribution statement

Wannida Sae-Tang: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis. **Ep Heuvelink:** Writing – review & editing, Visualization, Supervision, Formal analysis. **Céline C.S. Nicole:** Methodology, Investigation. **Elias Kaiser:** Writing – review & editing, Visualization, Methodology. **Kjell Sneeuw:** Methodology, Investigation. **Mexximiliaan M.S.F. Holweg:** Writing – review & editing, Investigation. **Sabrina Carvalho:** Methodology, Investigation. **Iris F. Kappers:** Writing – review & editing, Visualization, Methodology. **Leo F.M. Marcelis:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests. Leo Marcelis reports financial support was provided by Aphria RX

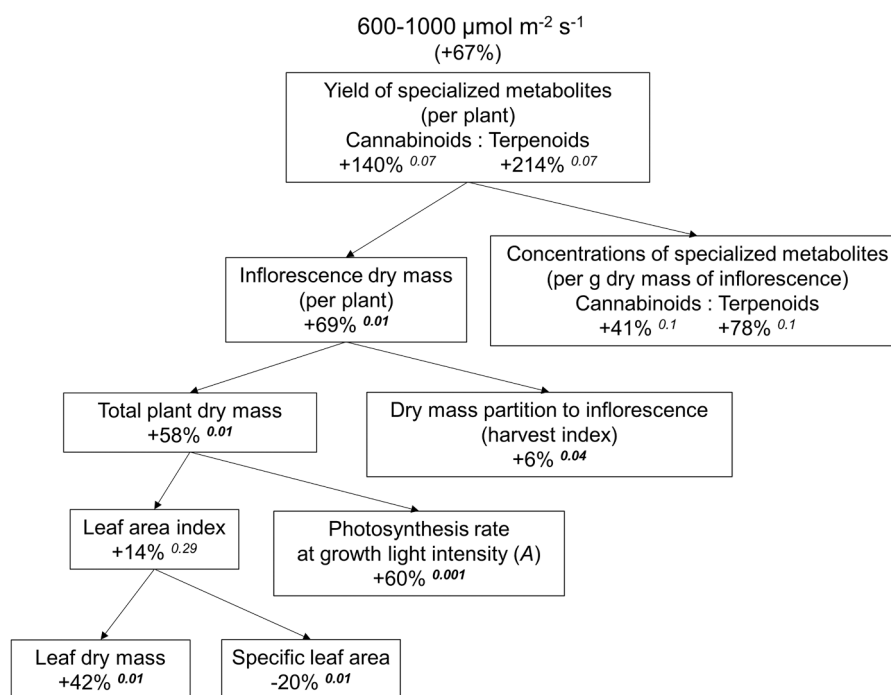


Fig. 7. Yield component analysis of the effects of light intensity during 8 weeks into the generative phase on specialized metabolites (IS response ratio on fresh mass basis) in medicinal cannabis. Percentages indicate the relative change in the components when PPFD increased from 600 to 1000 $\mu\text{mol m}^{-2}\text{s}^{-1}$, based on a linear regression (P-values indicated as superscripts) making use of the data at 600, 800 and 1000 $\mu\text{mol m}^{-2}\text{s}^{-1}$ (Table S2). The specialized metabolites, plant mass, and leaf area were measured after 8 weeks of generative phase. A represents net photosynthesis rate measured at growth light intensity (i.e. 600, 800, and 1000 $\mu\text{mol m}^{-2}\text{s}^{-1}$), 5 weeks into the generative phase.

GmbH, Germany. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jarmap.2024.100583](https://doi.org/10.1016/j.jarmap.2024.100583).

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